

Small test: I-type (add immediate), S-type(store word), R-type (add instructions)

10000093
00500113
00310193
00A00213
01400293
0020A023
0030A223
0000A303
00730393
00F00413
00100493
00840513
0040A583
00100613
00458693
00300713
00470793
00578813

```
C:\Users\craig\OneDrive\Documents\GitHub\GitHub\Semester-project-Computer-Architecture->python Comp_Arch_Project.py
HALT
steps = 34
final pc = 0x00000048
wrote trace.log, regs_final.log, dmem_final.log, cache.log, cache_stats.log
Number of stalls: 16

-----PERFORMANCE REPORT-----
Total cycles      : 34
Instructions      : 18
Total stalls      : 16
CPI               : 1.889

Stall Breakdown:
RAW-EX           : 4
RAW-MEM          : 6
RAW-WB           : 6

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```

Small fail test: I-type (addi dependency chain)

00100093
00108113
00110193
00118213
00120293
00128313
00130393
00138413
00140493

00148513

```
C:\Users\craig\OneDrive\Documents\GitHub\GitHub\Semester-project-Computer-Architecture->python Comp_Arch_Project.py
HALT
steps = 37
final pc = 0x00000028
wrote trace.log, regs_final.log, dmem_final.log, cache.log, cache_stats.log
Number of stalls: 27

-----PERFORMANCE REPORT-----
Total cycles      : 37
Instructions      : 10
Total stalls      : 27
CPI               : 3.700

Stall Breakdown:
  RAW-EX          : 9
  RAW-MEM         : 9
  RAW-WB          : 9

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```

Large test: I-type (addi and lw), S-type (save word), R-type (add/sub instructions)

10000093
00500113
00310193
00A00213
01400293
0020A023
0030A223
0000A303
00730393
00F00413
00100493
00840513
0040A583
00100613
00458693
00300713
00470793
00578813
00680893
00790913
00100993
00800A13
00A0AA93
0090AB13
00100B93
00400C13
00C0AC93

00B0AD13
00100D93
00E00E13
0000AE93
00D0AF13
00100F93
01001013
00F0A093
00E0A113
00101193
01201213
01302293
01402313
01503393
01603413
01704493
01804513
01905593
01A05613
01B06693
01C06713
01D07793
01E07813
01F08893
02008913
02109993
0220A013
0230AA93
0240AB13
0250BB93
0260BC13
0270BC93
0280BD13
0290BD93
02A0BE13
02B0BE93
02C0BF13
02D0BF93
02E0C013
02F0C093
0300C113
0310C193
0320C213
0330C293

0340C313
0350C393
0360C413
0370C493
0380C513
0390C593
03A0C613
03B0C693
03C0C713
03D0C793
03E0C813
03F0C893
0400C913
0410C993
0420CA13
0430CA93
0440CB13
0450CB93
0460CC13
0470CC93
0480CD13
0490CD93
04A0CE13
04B0CE93
04C0CF13
04D0CF93
04E0D013
04F0D093
0500D113
0510D193
0520D213
0530D293
0540D313
0550D393
0560D413
0570D493
0580D513
0590D593
05A0D613
05B0D693
05C0D713
05D0D793
05E0D813
05F0D893

0600D913
0610D993
0620DA13
0630DA93
0640DB13
0650DB93
0660DC13
0670DC93
0680DD13
0690DD93
06A0DE13
06B0DE93
06C0DF13
06D0DF93
06E0E013
06F0E093
0700E113
0710E193
0720E213
0730E293
0740E313
0750E393
0760E413
0770E493
0780E513
0790E593
07A0E613
07B0E693
07C0E713
07D0E793
07E0E813
07F0E893
0800E913
0810E993
0820EA13
0830EA93
0840EB13
0850EB93
0860EC13
0870EC93
0880ED13
0890ED93
08A0EE13
08B0EE93

08C0EF13
08D0EF93
08E0F013
08F0F093
0900F113
0910F193
0920F213
0930F293
0940F313
0950F393
0960F413
0970F493
0980F513
0990F593
09A0F613
09B0F693
09C0F713
09D0F793
09E0F813
09F0F893

```
C:\Users\craig\OneDrive\Documents\GitHub\GitHub\Semester-project-Computer-Architecture->python Comp_Arch_Project.py
HALT
steps = 212
final pc = 0x000002C8
wrote trace.log, regs_final.log, dmem_final.log, cache.log, cache_stats.log
Number of stalls: 34

-----PERFORMANCE REPORT-----
Total cycles      : 212
Instructions      : 178
Total stalls      : 34
CPI               : 1.191

Stall Breakdown:
RAW-EX   : 10
RAW-MEM  : 12
RAW-WB   : 12

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```

Big test fail(addi dependency chain)

00108093

00108093

00108093

00108093

00108093

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00108093

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00108093

00108093

00108093

[illegible]

[illegible]


```
C:\Users\craig\OneDrive\Documents\GitHub\GitHub\Semester-project-Computer-Architecture->python Comp_Arch_Project.py
HALT
steps = 801
final pc = 0x00000324
wrote trace.log, regs_final.log, dmem_final.log, cache.log, cache_stats.log
Number of stalls: 600

-----PERFORMANCE REPORT-----
Total cycles      : 801
Instructions      : 201
Total stalls      : 600
CPI               : 3.985

Stall Breakdown:
  RAW-EX  : 200
  RAW-MEM : 200
  RAW-WB  : 200

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```